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SECTION 02

PROJECT (FINAL REPORT)

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1. INTRODUCTION

1.1 Project Title

Retails Sales Performance and Customer Purchasing Behaviour Analysis.

1.2 Problem Background

<u>Business Context</u>

Pricing, product availability and the demand in various regions are constantly changing in the retail business. Being able to process these changes effectively is often difficult for retailers who have relied on manual or siloed data review processes for their data. If they don't have a single analytical system, these key indicators are siloed in disjointed data sets — like low-performing product categories, seasonal demand fluctuations, or discount rates that aren't leading to volume uplift — and thus hidden from view.

This project aims to fill in the gap by creating a Business Intelligence solution to a multi-categorized retail environment. The solution integrates three data sets containing product pricing and product demand data signals, a product catalog and historical sales transactions for 2023 to 2024 which can then be correlated by analytical analysis to provide an analysis not possible using any one of these data sets alone.

<u>Importance of the Problem</u>

There is a direct correlation between pricing strategy and the sensitivity of demand and the profitability. A retailer with uniform (same percentage) discounts, regardless of tier, without the knowledge of whether the discounts are working or not is destroying margins without getting a higher revenues return. Likewise, a retailer with no insight into the seasonality of revenues will not be able to optimize the investment in his inventory, staffing or marketing campaigns according to the seasons in which he will create the greatest value from his business. With no structured BI pipeline, decision makers are forced to make decisions based on intuition or on late reporting, which puts their operations at risk.

<u>Expected Impact</u>

It creates an interactive multi-dimensional dashboard which will allow retail stakeholders to see where the revenue is getting concentrated (by tier, category and region), to assess the return on a promotion based on a discount, and to plan around verified seasonal demand. The expected outcomes are to make more informed pricing decisions, allocate marketing spend optimally and to have a data-driven basis for category management and a regional growth strategy.

1.3 Project Objectives

The following 5 objectives will be used to design and evaluate this project:

1. To create and develop an end-to-end Business Intelligence solution that integrates retail pricing signals, product catalog metadata and sales history into a single analytical pipeline.
2. To create an Alteryx Designer workflow to cleanse, transform and engineer the features of all three datasets to be ready for downstream visualisation and combine the datasets.
3. To determine if there are any "spikes" or "dips" in revenue throughout the year, and which revenue products are successful or not, based on the sales trends over the course of 2023-2024 and how they correlate with the demand signals in each product category.
4. To create an interactive Power BI dashboard to enable stakeholders to investigate by product tier, determine slow moving stock and benchmark by product category and region.
5. To gain business insights and recommendations from the visualised data that will help improve business solutions to enhance retail sales strategy in the future.

1.4 Dataset Description

1.4.1 Retails Pricing & Demand Signals Dataset (DS1)

- **Source**: Kaggle –Retails Pricing & Demand Signals Dataset
- **Type of Data**: Structured
- **Shape**: 172,800 rows x 17 columns

This data provides price observation and demand data at a daily level for products over a variety of regions, channels and seasons. It is used as a main source for pricing analysis, evaluation of the effectiveness of discounts, and modelling of demand signals in this project.

Table 1.4.1 Retails Pricing & Demand Signals Dataset

Column name	Data type	Description
date	Object (String)	Date of the pricing and demand observation
product_id	Object (String)	Unique identifier for the product
category	Object (String)	Product category
brand	Object (String)	Product brand
region	Object (String)	Geographic market or region
channel	Object (String)	Sales channel such as web, mobile, or app
season	Object (String)	Seasonal period associated with the date
base_price	Float64 (Decimal)	Original product price before promotion or discount
current_price	Float64 (Decimal)	Effective selling price after discount

price_change_pct	Float64 (Decimal)	Percentage change from base price
discount_pct	Int64 (Integer)	Discount percentage applied to the product
promotion_type	Object (String)	Type of promotion applied
units_sold	Int64 (Integer)	Number of units sold during the observation period
revenue	Float64 (Decimal)	Total revenue generated from units sold
inventory_level	Int64 (Integer)	Available inventory after sales activity
stockout_flag	Boolean	Indicates whether inventory is critically low
demand_index	Float64 (Decimal)	Normalized demand score for comparison across products

1.4.2 Retail Product Catalog Dataset (DS2)

- **Source**: Kaggle –Retail Product Catalog Dataset (noopurbhatt)
- **Type of data**: Structured
- **Shape**: 20 rows x 11 columns

This is a final dataset that is used as a product reference catalog with metadata for every product such as the type of product, brand, price, ratings and availability. It's attached to DS1 at the common. This product_id key will be used to augment the pricing data set and add product-level information from the catalog.

Table 1.4.2 Retail Product Catalog Dataset

Column name	**Data type**	**Description**
product_id	Int64 (Integer)	Unique numeric identifier assigned to each product in the catalog
product_name	Object (String)	Full descriptive name of the product as listed in the retail catalog
brand	Object (String)	Manufacturer or brand name associated with the product
category	Object (String)	Top-level product category (e.g., Groceries, Electronics, Clothing, Home & Kitchen, Personal Care, Sports & Outdoors)
sub_category	Object (String)	More specific classification within the main category (e.g., Dairy, Audio, Skincare, Fitness, Cookware)
price	Float64 (Decimal)	Listed retail price of the product in USD (ranges from $3.49 to $599.99)
rating	Float64 (Decimal)	Average customer rating of the product on a scale of 1.0 to 5.0

review_count	Int64 (Integer)	Total number of customer reviews submitted for the product
stock_status	Object (String)	Current inventory availability status of the product (e.g., In Stock, Out of Stock)
tags	Object (String)	Pipe-separated keywords associated with the product used for search and recommendation purposes
description	Object (String)	Short textual description summarising the product features and specifications

1.4.3 Product Sales Dataset 2023-2024 (DS3)

- **Source**: Kaggle –Product Sales Dataset 2023-2024 (yashyennewar)
- **Type of data**: Structured
- **Shape**: 200,000 rows x 14 columns

The data set includes historical sales transaction data from January 2023 to December 2024 from a variety of U.S. regions, states and product categories. It serves as the basis for analyzing revenue trends, comparing seasonal unit performance and benchmarking against other regions.

Table 1.4.3 Product Sales Dataset 2023-2024

Column name	**Data type**	**Description**
Order_ID	Int64 (Integer)	Unique numeric identifier assigned to each sales order transaction
Order_Date	Object (String)	Date on which the order was placed, covering the period from January 2023 to December 2024
Customer_Name	Object (String)	Full name of the customer who placed the order
City	Object (String)	City in which the customer or delivery address is located
State	Object (String)	US state corresponding to the customer or delivery address
Region	Object (String)	Geographic US region of the order (South, Centre, East, West)
Country	Object (String)	Country of the order destination (United States)

Category	Object (String)	Top-level product category of the sold item (Accessories, Clothing & Apparel, Electronics, Home & Furniture)
Sub_Category	Object (String)	Specific sub-category within the broader product category (e.g., Smartphones, Sportswear, Furniture, Kitchenware)
Product_Name	Object (String)	Name of the specific product associated with the order line
Quantity	Int64 (Integer)	Number of units ordered in the transaction (ranges from 1 to 11)
Unit_Price	Float64 (Decimal)	Price per unit of the product at the time of sale (ranges from $17.03 to $1,432.00)
Revenue	Float64 (Decimal)	Total revenue generated from the order line (Quantity x Unit_Price), ranging from $17.03 to $9,014.25
Profit	Float64 (Decimal)	Net profit earned from the order line after costs, ranging from $3.92 to $2,763.72

1.5 Proposed BI Approach

Alteryx DesignerWorkflow

The entire data preparation pipeline will be created using Alteryx Designer. There are 5 stages to the workflow:

- **Data Input**: All 3 CSV files are loaded with Input Data tools. Each input is followed by a Select tool to ensure field types are consistent, to rename the fields for consistency across the datasets and to remove unused fields.
- **Data Cleansing**: Each of the datasets is processed through a Data Cleansing tool, a Unique tool and a Filter tool (Data Cleaning). The promotion_type is filled with the promotion label ‘None' (58% of DS1 rows) as the absence of such promotion can be a valid state. All string fields get whitespace removed from them. There are no duplicates on dataset specific key fields. The rows that have no revenue and no quantity or have an invalid quantity are removed. A Formula tool is used for extracting the month because the DateTime tool was not able to reliably extract the month for DS3, where the Order_Date is stored in MM-DD-YYYY format.
- **Data Transformation**: Using a Formula tool, seven new analytical columns are created: month, year, discount_amount, price_sensitivity_flag, stock_status, revenue_per_unit, and product_tier. Then, a Summarise tool is used to pull out the data by category, season, region, month, year and product tier to calculate SUM(revenue), SUM(units_sold), AVG(demand_index), and COUNT(product_id).
- **Data Integration**: DS1 and DS2 are joined on product_id, using a left join so that all the pricing data in DS1 is retained. The category labels are then standardised across all datasets using a Formula tool, and then a second Left Join is used to join this merged table with the DS3 on the category level. For categories like Beauty, Sports and Groceries (which do not have an equivalent in DS3), a Union tool is used to pass all the 172,791 records through the matched (J) and unmatched (L) output ports of Join 2.
- **Output**: The final cleaned, transformed and integrated data will be exported as a CSV file for use in Power BI.

Visualization Approach

The two interactive dashboards and one storyboard were created using Microsoft Power BI Desktop. The following five chart types are in order of planning:

- **KPI Cards**: Summarize Total Revenue, Total Units Sold, Revenue per Unit and Product Count at a glance to instantly give the business context when the dashboard loads.
- **Clustered Bar Chart**: This chart displays the revenue for each product tier (Budget, Mid-Range, Premium) for each month and shows the month-over-month growth or decline, plus the amount of revenue generated by each tier of each month.
- **Line Chart**: Continuous trend line of total revenue makes it easy for the stakeholder to see when the revenue is highest or lowest in a month or season.
- **The Matrix / Heatmap Table**: allows the user to present a cross tabulation of revenue by region (rows) and category (columns), which can be quickly used to identify the strongest and weakest combinations of region/category.
- **Donut Chart**: Represents the distribution of revenue between Winter and Spring, and indicates the proportionate revenue imbalance between these seasons in an easy-to-understand way for non-technical stakeholders
- **Scatter Chart**: visualises the relationship between discount level and demand index for each product tier, directly showing if there is a relationship between depth of discount and tiers demand response.

To facilitate interactive filtering and cross-dimensional exploration, all dashboards have Region, Category and Product Tier slicers. This storyboard tells the analytical story from the KPI overview to the category comparison and then the seasonality insight, guiding the interpretation of the data for presenting.

2. METHODOLOGY & WORKFLOW

This part mentions about the methodology chosen for this project, and the overall Alteryx Designer workflow done to make this project a success.

2.1 Overview of the Workflow Approach

The methodology for this project was carried out entirely in Alteryx Designer, covering the full data preparation pipeline from raw input to a single export-ready dataset. The workflow integrates three structured datasets sourced from Kaggle which are the Retail Pricing & Demand Signals dataset acting as the base table, the Retail Product Catalog dataset enriching it with product metadata, and the Product Sales dataset contributing transaction-level revenue figures. Each dataset was prepared on its own branch before being merged, so an issue found in one source could be corrected without disturbing the others. The process followed the BI architecture set out in Phase 1, which structures the pipeline into five layers including data source, data preparation, data storage, visualization, and decision support. This staged branch-then-merge structure follows standard ETL design practice, where data is extracted and cleaned independently per source before being transformed and integrated into a unified structure (Fana et al., 2021).

2.2 Data Integration Process

Two sequential join operations were used to bring the three datasets together rather than attempting a single three-way merge, as staging the joins keeps every step traceable and makes it easier to isolate errors.

Join 1 merged the Retail Pricing & Demand Signals dataset (DS1) with the Retail Product Catalog (DS2) on product_id, using a Left join so that no DS1 records would be dropped even where no catalog match existed, as shown in Figure 2.1. A select tool was placed immediately after this join to remove duplicated columns coming from DS2, keeping the schema clean going into the next step.

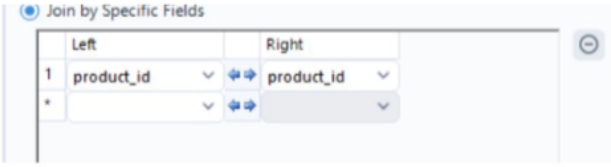


Figure 2.1 *Join 1 configuration panel, join condition: DS1.product_id=DS2.product_id*

Before the second join could happen, the category labels across all three datasets needed to be standardized, since each dataset was authored independently on Kaggle and used different category vocabularies. A formula tool handled this remapping, leaving categories with no DS3 equivalent unchanged, which is the expected outcome under a Left join, as shown in Figure 2.2.

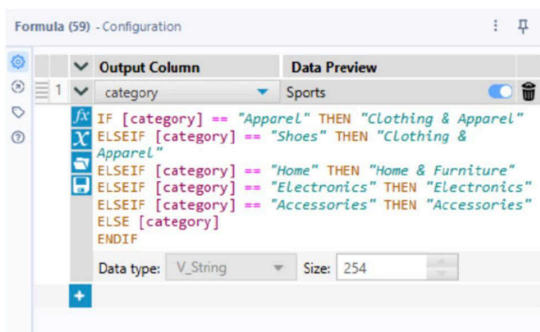


Figure 2.2 *Formula tool showing the category standardization expression*

The Product Sales dataset (DS3) is order-level data with over 200,000 rows, so joining it directly at the product level would have caused row multiplication. To avoid this, DS3 was pre-aggregated using a Summarise tool, grouped by category and order month,with Revenue and Quantity summed into total_revenue and total_quantity. This reflects a core principle in dimensional modelling, where fact tables being joined must share a consistent grain, or row meaning. Aggregating DS3 from the order level to the category-month level before the join brought it to the same grain as the DS1+DS2 table, avoiding the row multiplication that joining mismatched grains would otherwise cause (Kimball & Ross, 2013).

Join 2 then merged the DS1+DS2 table with this aggregated DS3 table on the standardized category field, again using a Left join to preserve all records, as shown in Figure 2.3. Following the join, a Union tool combined the matched (J) and unmatched (L) output ports, ensuring all 172,791 records carried through to the next stage rather than being silently dropped.

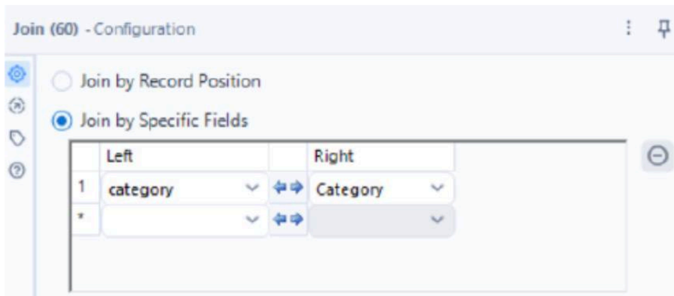


Figure 2.3 *Join 2 configuration panel, join condition: merged.category=DS3_agg.Category*

2.3 Data Cleaning Process

Cleaning was performed before integration, on each dataset's own branch, following the principle of cleaning as early in the pipeline as possible so downstream joins and aggregations would operate on consistent, analytically meaningful data.

A consistent three-tool cleaning pipeline was applied to each dataset which are a Data Cleansing tool to handle blanks and whitespace, a Unique tool to remove duplicate records, and a Filter tool to exclude invalid rows. These tools followed directly after Member 1's Select tools, which had already confirmed and corrected each field's data type.

The pre-cleaning audit found that the most significant data quality issue was in DS1's promotion_type column, where 100,155 of 172,800 rows (58%) were null. Rather than dropping these rows, the Data Cleansing tool replaced the nulls with the label "None", since the absence of a promotion is a valid and expected state, not missing data, as shown in Figure 2.4. The Unique tool was keyed on date + product_id + region for DS1, and the audit confirmed all 172,800 rows were already unique combinations on that key. The Filter tool then removed 9 rows representing zero-sales events ([units_sold] >= 0 AND [revenue] > 0), bringing DS1 down to 172,791 clean rows.

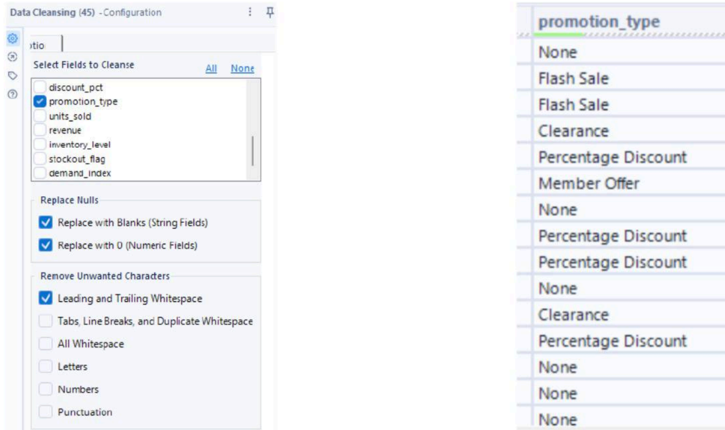


Figure 2.4 *Data Cleansing configuration for promotion_type, and resulting null replacement with 'None'*

DS2 (the Product Catalog) and DS3 (Product Sales) were both already clean on inspection, with no nulls, duplicates, or invalid values found across their respective 20 and 200,00 rows. The same Data Cleansing and Unique tools were still applied to both as a structural safeguard, protecting the pipeline if the source files were ever updated upstream.

2.4 Data Transformation Process

Once the datasets were integrated, a Formula tool was used to derive seven new columns that the Phase 3 dashboards are built directly around, meaning no further calculation was needed once the data reached Power BI. These included month and year, discount_amount, price_sensitivity_flag, stock_status, revenue_per_unit, and product_tier.

After the derived columns were created, a Summarise tool collapsed the merged table from roughly 172,000 rows into a grouped summary by category, season, region, month, year, and product tier, computing sums, averages, and a product count per group, as shown in Figure 2.5. The final aggregated output was exported using the Output Data tool as SkinAqua_Final_Output.csv, then opened briefly in Excel to sanity-check that all derived fields and category labels behaved as expected.

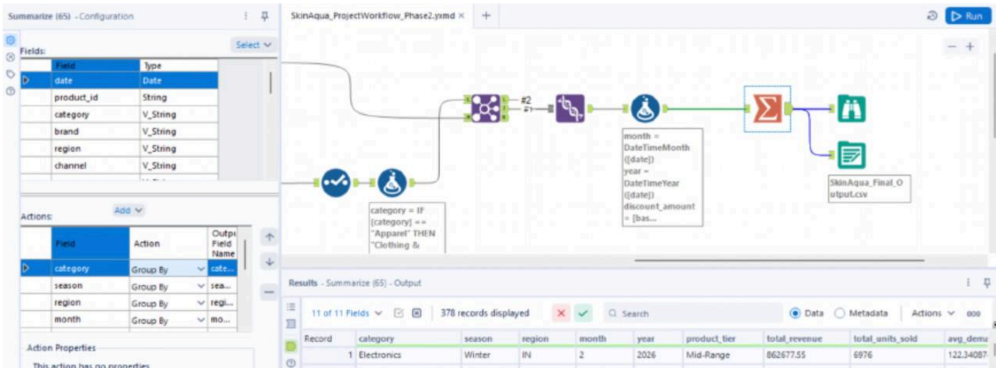


__Figure 2.5__ Summarise and Output Data tool configuration, final group-by fields, aggregations, and CSV export

2.5 Results of the Workflow

Running the complete workflow, illustrated end-to-end in Figure 2.6, produced a Browse tool output confirming 172,791 records and 21 fields, as shown in Figure 2.7. The field count reflects DS1's 17 original columns plus the four fields contributed by DS3's aggregation. After the final Summarise step, the exported SkinAqua_Final_Output.csv contained 378 aggregated rows, ready to be picked up directly in Power BI for Phase 3, with every derived column needed for the dashboards already in place.

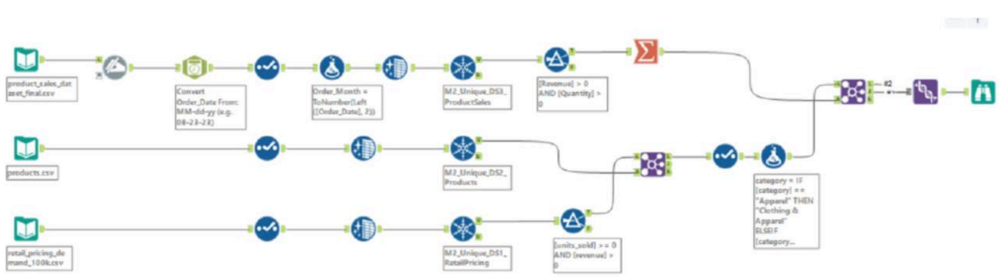
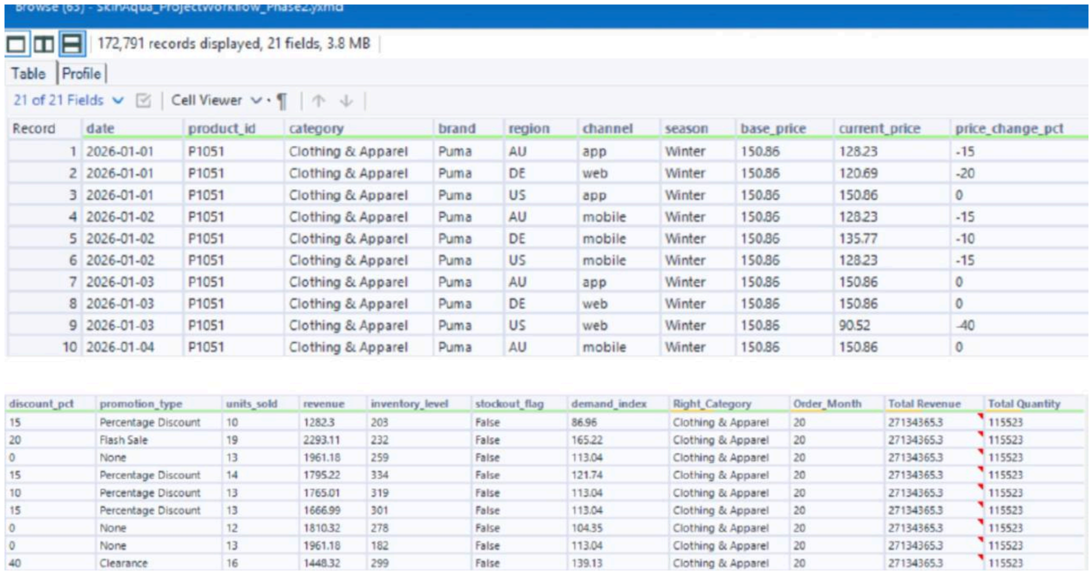


__Figure 2.6__ Data flow diagram: DS1 → Join 1 ← DS2 → Select → Formula (category) → Join 2 ← DS3 (Formula + Summarise) → Union → Browse



172,791 records displayed, 21 fields, 3.8 MB

Table | Profile

21 of 21 Fields | Cell Viewer

Record	date	product_id	category	brand	region	channel	season	base_price	current_price	price_change_pct
1	2026-01-01	P1051	Clothing & Apparel	Puma	AU	app	Winter	150.86	128.23	-15
2	2026-01-01	P1051	Clothing & Apparel	Puma	DE	web	Winter	150.86	120.69	-20
3	2026-01-01	P1051	Clothing & Apparel	Puma	US	app	Winter	150.86	150.86	0
4	2026-01-02	P1051	Clothing & Apparel	Puma	AU	mobile	Winter	150.86	128.23	-15
5	2026-01-02	P1051	Clothing & Apparel	Puma	DE	mobile	Winter	150.86	135.77	-10
6	2026-01-02	P1051	Clothing & Apparel	Puma	US	mobile	Winter	150.86	128.23	-15
7	2026-01-03	P1051	Clothing & Apparel	Puma	AU	app	Winter	150.86	150.86	0
8	2026-01-03	P1051	Clothing & Apparel	Puma	DE	web	Winter	150.86	150.86	0
9	2026-01-03	P1051	Clothing & Apparel	Puma	US	web	Winter	150.86	90.52	-40
10	2026-01-04	P1051	Clothing & Apparel	Puma	AU	mobile	Winter	150.86	150.86	0

discount_pct	promotion_type	units_sold	revenue	inventory_level	stockout_flag	demand_index	Right_Category	Order_Month	Total Revenue	Total Quantity
15	Percentage Discount	10	1282.3	203	False	86.96	Clothing & Apparel	20	27134365.3	115523
20	Flash Sale	19	2293.11	232	False	165.22	Clothing & Apparel	20	27134365.3	115523
0	None	13	1961.18	259	False	113.04	Clothing & Apparel	20	27134365.3	115523
15	Percentage Discount	14	1795.22	334	False	121.74	Clothing & Apparel	20	27134365.3	115523
10	Percentage Discount	13	1765.01	319	False	113.04	Clothing & Apparel	20	27134365.3	115523
15	Percentage Discount	13	1666.99	301	False	113.04	Clothing & Apparel	20	27134365.3	115523
0	None	12	1810.32	278	False	104.35	Clothing & Apparel	20	27134365.3	115523
0	None	13	1961.18	182	False	113.04	Clothing & Apparel	20	27134365.3	115523
40	Clearance	16	1448.32	299	False	139.13	Clothing & Apparel	20	27134365.3	115523

Figure 2.7 *Browse tool output confirming 172,791 records and 21 fields*

2.6 Tools Used in the Workflow

Table 2.1 below summarizes the Alteryx tools used throughout the workflow and the specific role each one played.

Table 2.1 *Alteryx tools used in the workflow*

Tool	Role in the Workflow
Input Data	Loads the three source CSV files
Select	Selects fields and corrects data types
Dynamic Rename	Trims whitespace and standardizes text case
DateTime	Converts Order_Date text to a date field
Date Cleansing	Trims whitespace and standardizes text case
Unique	Removes duplicate records per branch
Filter	Removes invalid records

Formula	Standardizes category labels; creates derived columns
Summarize	Aggregates sales to category/month level, the final group-by
Join	Integrates the datasets on product_id and category
Union	Consolidates the integration output
Browse	Inspects and verifies the final dataset
Output Data	Writes the integrated dataset to CSV for visualization

2.7 Challenges Faced During the Workflow

A few data quality and structural issues came up during the build, all resolved within the workflow itself:

A. Product_id type and range mismatch

DS1 stored product_id as a string while DS2 stored it as an integer. Recasting DS2's product_id to string in the Select tool prevented a type error, but the numeric ranges remained incompatible, so Join 1 returned zero matched rows. All DS1 records exited via the Left port with DS2 columns null.

B. Category naming inconsistency

Each dataset used a different category vocabulary, with only "Electronics" matching exactly across DS1 and DS3. A Formula tool remapped DS1's categories to DS3's naming convention ahead of Join 2.

C. Whitespace in DS3 column names

Fields like 'Revenue' and 'Unit_Price' carried leading spaces that would have caused silent reference errors. A Select tool renamed these fields before they were used in the Summarise tool.

D. DS3's date format

Order_Date was stored as MM-DD-YYYY rather than the standard format used in DS1, and the DateTime tool couldn't reliably parse the two-digit year, so month extraction was done manually using Left([Order_Date],2) in a Formula tool.

E. Categories with no DS3 match

Beauty, Sports, and Groceries had no equivalent category in DS3, so these rows returned null DS3 columns after Join 2. A Union tool combining the matched and unmatched join outputs ensured these records were retained rather than dropped.

3. VISUALIZATION & INSIGHTS

There are two dashboards that have been designed. The first dashboard is about Sales KPI & Trend Overview, and the second dashboard is about comparison and design.

3.1 Dashboard 1 (Sales KPI & Trend Overview)

As shown in Figure 3.1 is the first dashboard about sales KPI and trend overview. The goal of this dashboard is for the user to know the current total revenue, total units sold, revenue per unit, product count, comparison between the total revenue by month and product tier, and trends of the total revenue by month. This dashboard could be viewed by selecting the region, category, and product tier filters. The goals in every chart in the developed dashboard is described further in detail as below:

1. Dashboard header (Sales KPI & Trend Overview title)

   Goal: Provide rough context and orientation for the viewer so they know what the dashboard is about before reading any data.

   Justification: Text box is correct choice because it did not carry any data, instead it just provide an overview of the dashboard.

   Best practice: Position in the top-left quadrant follows the F-pattern reading flow. It reserves space for the report title and subtitle without competing with data visuals.

   Interactivity: Static, as it acts as a stable reference point only.

2. Dropdown Slicers (Region, Category, Product Tier)

   Goal: Allow users to filter the entire project page by choosing three key business dimensions. All three slicers simultaneously update all other visuals on the page.

   Justification: Dropdown mode in the slicers is the right choice because not all values need to be permanently visible. Placing them in the top-right corner (header zone) follows the convention that filters live above/beside the content they control, not mixed into the chart area.

Best practice: Grouping all slicers in the same row creates a single filter strip, reducing cognitive overhead. The dropdown format also prevents the from dominating layout space.

Interactivity: All three use isInvertedSelectionMode, meaning the default state shows all data and clicking a value excludes everything else. This encourages users to start broad and narrow down.

3. KPI cards (Total revenue, Units sold, Revenue per Unit, Product Count)

   Goal: To deliver the most important headline metrics at a glance before the user digs into charts. These act as the executive summary row.

   Justification: Card visuals are the gold standard for KPIs because they remove all chart overhead and put the metric front and centre. Using four cards in horizontal row creates a scannable summary band and viewed in a correct hierarchy (filters → summary → detail)

   Best practice: The alignment is centered, consistent font size, and all in the same dimensions, as this visual symmetry signals they are peers of equal importance.

   Interactivity: As drillFilterOtherVisuals: true is set, clicking any of the card values acts as a filter signal to other visuals. The cards also respond to slicer selections where filtering by region, or tier, all four KPI values update dynamically.

4. Clustered Column Chart (Revenue by Month x Product Tier)

   Goal: To answer questions such as "Which product tier is driving revenue each month, and how does that compare across months?". It shows trend analysis and tier level composition analysis in a single chart.

   Justification: A clustered column chart is chosen because it has two categorical dimensions and group, and wants to compare values across months. The clustered column gives both comparison simultaneosly.

   Best practice: Each tiers gets its own coloured bar per month, making the legend actionable. The x-axis is labelled "Month" explicitly via the titleText property, following the rule that axes should always be identified

Interactivity: By clicking any bar on the chart cross filters the line chart and update the KPI cards. For example, clicking the "Premium" bar for March would filter all other visuals to show only Premium tier data for March

5. Line chart (Revenue over Month)

   Goal: To show the overall revenue trajectory across months and answer questions such as "what is the shape of our revenue over time?"

   Justification: Line chart could display the trend between them as the line chart is the best choice for continuous time-series data.

   Best Practice: Months is on the X-axis, revenue on the Y-axis, and a single smooth line connects the data points.

   Interactivity: By clicking a data point on the line chart (a specific month) cross-filters the column chart and all KPI cards to that month. Combined with the slicer filters, a user can, select "East region + Premium tier" via slicers and then click April on the line chart to see exactly what Premium tier revenue looked like in East region in April across all visuals simultaneously.

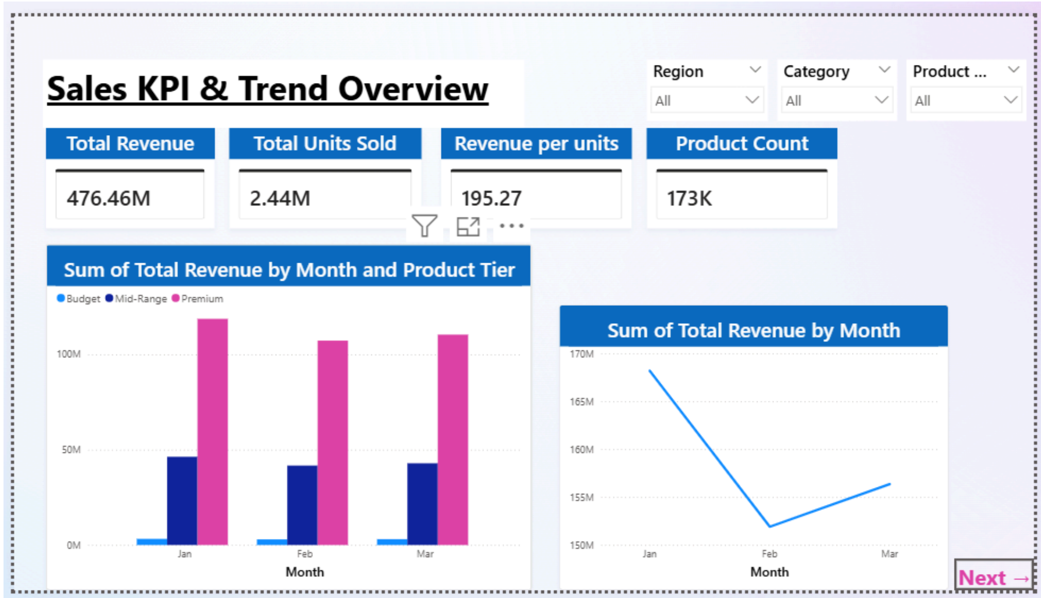


Figure 3.1 Dashboard 1

3.2 Dashboard 2 (Comparison and Insights)

Dashboard 2 compares performance across product categories, regions, seasons, and discount strategy as shown in Figure 3.2.

1. Combo Chart (Revenue vs Demand by Category)

   Goal: To reveal mismatches: categories with high revenue but low demand (potential saturation) or low revenue but high demand (untapped opportunity)

   Justification: A combo chart is suitable when two metrics share the same x-axis but have different units and scales. Revenue is plotted as bars on the primary Y-axis, while the Avg Demand Index is plotted as a line on the secondary Y-axis. Using two bar series side-by-side would conflate the scales; two line charts would lose the revenue magnitude.

   Best practice: The two series are differentiated by both chart type (bar vs line) and colour, ensuring they are distinguishable even in greyscale. Labels on the bars remove the need for the viewer to read the y-axis precisely.

Interactivity: Clicking any bar or line data point cross-filters other visuals on the page.

2. Donut Chart (Revenue Share by Season)

   Goal: To show the proportional split of total revenue between two seasons (Spring and Winter) to surface whether it has a seasonal skew.

   Justification: A donut chart is appropriate for a simple part-of-whole comparison with very few categories (just 2 here). It is preferred over a full pie chart because the hollow centre allows a key callout label to sit inside, reducing the need for external annotation.

   Best practice: he two segments use visually distinct and meaningfully contrasting colours

   Interactivity: Clicking a segment cross-filters all other visuals on the page.

3. Matrix Table Heatmap (Region × Category)

   Goal: To answer "which region-category combination generates the most revenue?" across all six regions and all product categories simultaneously.

   Justification: Both dimensions are categorical and the audience needs exact numeric values alongside the colour pattern.

   Best practice: Row and column totals are included, following the standard practice for cross-tabular business reports.

   Interactivity: Clicking a cell cross-filters other visuals to that specific region-category combination.

4. Bubble Chart (Discount Rate vs Demand by Product Tier)

   Goal: Explore the relationship between discount rate (%) and average demand index, segmented by product tier

Justification: A bubble chart is the only standard chart type that can simultaneously encode three continuous variables

Best practice: The x-axis is labelled "Discount Rate %" and the y-axis shows demand, both with grid lines for accurate reading.

Interactivity: Cross-filtering is active where clicking a bubble (tier) filters the heatmap, combo chart, and donut simultaneously to that tier.

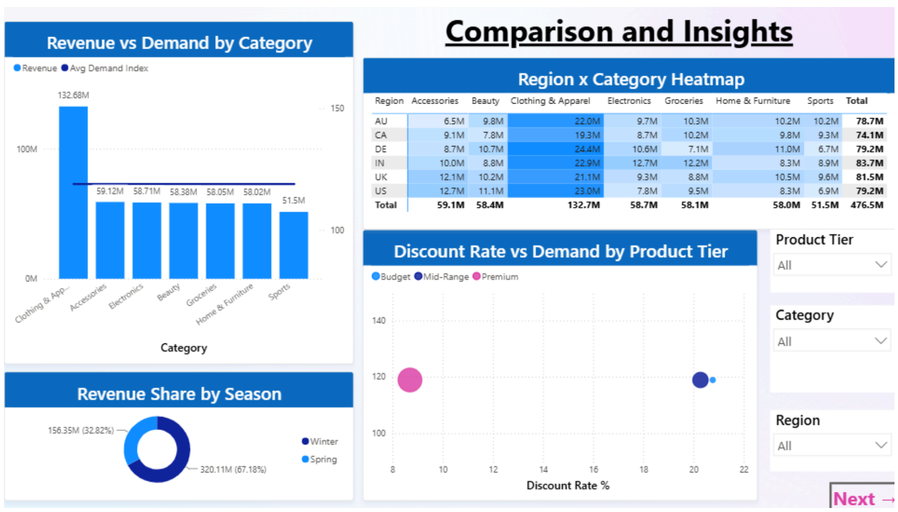


Figure 3.2 Dashboard 2

4. DISCUSSION

This project successfully delivered an end-to-end Business Intelligence solution integrating retail pricing signals, product catalog metadata, and historical sales transactions into a single analytical pipeline. The combined use of Alteryx Designer for data preparation and Microsoft Power BI for interactive visualization enabled a structured and reproducible approach to retail sales analysis that would not have been possible from any single dataset alone.

The analytical findings across both dashboards highlight several commercially significant patterns. The identification of discount inefficiency across Budget and Mid-Range product tiers is particularly noteworthy, as the data clearly shows that heavier discounting is not generating proportionally higher demand, which represents a direct margin risk for the business. This insight would be difficult to surface without cross-referencing discount rates, demand indices, and revenue figures simultaneously, which is precisely the kind of multi-dimensional analysis a structured BI pipeline enables.

The revenue concentration in Clothing & Apparel and in the Winter season similarly points to strategic vulnerabilities that stakeholders should address proactively. A business that depends heavily on a single category and a single seasonal period is exposed to significant downside risk if either declines. The dashboards make these concentrations immediately visible to non-technical stakeholders through the heatmap and donut chart visualizations, lowering the barrier for data-driven decision-making at the management level.

From a technical standpoint, the project reinforced the importance of data quality as a prerequisite for reliable analysis. Several data issues identified during the Alteryx workflow, including product ID type mismatches, inconsistent category labelling across datasets, and a flawed discount rate calculation, had downstream effects on the accuracy of certain dashboard metrics and required careful correction before the visualizations could be trusted. This underscores that the ETL phase is not merely a preparatory step but a critical determinant of analytical validity.

Looking ahead, the solution could be strengthened by incorporating a broader date range beyond the current three-month window (January to March), which limits the confidence of trend observations. Expanding the product catalog to full inventory depth would also enable product-level pricing analysis rather than category-level aggregation. If integrated with a forecasting model, the monthly revenue trend data could generate forward-looking demand signals to support proactive planning rather than retrospective review alone.

5. CONCLUSION

The successful implementation of the full Business Intelligence solution for retail sales performance and customer purchasing behaviour analysis proved to be successful in this project. The project consisted of three datasets with more than 372,000 records, combined into one analytical pipeline using an Alteryx Designer workflow and presented into two interactive Power BI dashboards that provided a unified view of the data and business.

All five of the project's objectives were achieved. An end-to-end solution of a BI architecture with data input, cleaning, transformation, integration and visualisation was implemented. The workflow addressed a significant data quality challenge within the pricing data set: 58% of the promotion type was null, and directly contributed to the dashboard analysis with seven new features that were created using the data sets. The interactive visualisations allowed for the trending, comparison, and KPIs analysis across product tier, product category, region, and season dimensions.

The most crucial analytical finding of this project is that even though the products are of a premium tier, they still don't have the highest discount rates because of which the revenue generated from the Clothing & Apparel is more than double of each and every region in comparison with other competitive categories. These results together indicate that the store has a few high-dollar items and one strong category; a commercial success and significant strategic vulnerability if the demand in either category declines. The seasonal analysis further shows that Winter (67.18% of revenue, or $320.11M) is the key to the annual result, with February being a month with a significant operational pulldown which should be considered in identifying areas for focus when planning the operational season.

The project itself was a technical success and was a means of reinforcing the value of having a structured BI pipeline. By enforcing a reproducible, auditable process for data-cleaning and transforming it, the use of Alteryx Designer for data preparation minimized manual handling and guaranteed that the data that went into Power BI was consistent, correctly typed and provided with analytical features. By using the interactive slicers in both dashboards, users were able to recombine data across dimensions, on-demand, adding to the analytical value of the fixed visuals.

We hope that, for future work, several improvements can be found which makes this solution more analytical and more useful in practice. If the product catalog was expanded to a full product inventory, then it would be possible to analyze product level prices instead of category level prices. If customer demographic information could be added, customer segmentation could go beyond tier and region to buying behaviour patterns. If a time-series forecasting model is applied to the monthly revenue data, the forecast will give forward-looking signals of demand, not just a summary of past revenues. Further, if the Alteryx workflow could be automated to refresh with live data sources, then this proof-of-concept could become an operational BI tool that would enable real-time business decisions to be made.

Finally, the project shows that the derived value of any multi-source data from any open data portal can be commercially valuable if channeled through an effective BI pipeline. Alteryx's data engineering tooling and Power BI's interactive visualisation were effective in identifying trends or patterns in pricing, demand, geography and seasonality that wouldn't have appeared in any one dataset alone.

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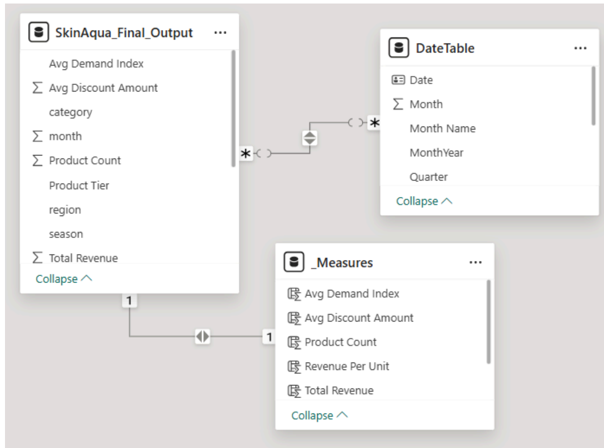
APPENDIX



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Avg Discount Amount
category
month
Product Count
Product Tier
region
season
Total Revenue
Collapse
DateTable
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Avg Discount Amount
Product Count
Revenue Per Unit
Total Revenue
Collapse